20:1 Micro Cycloidal Drive - Overview & Instructions

Product Description

The 20:1 Micro Cycloidal Drive is a compact, high-precision gear reduction solution designed for desktop robotics, hobbyist mechanical builds, and light industrial automation. This drive delivers exceptional high torque density in a small package, while providing smooth operation, excellent backlash performance, and superior shock absorption capabilities. Perfect for applications where precision and durability matter in a space-efficient form factor.

What's Included

Standard Package (STL Files)

- Cycloidal Discs:
- 9 size variations:
- 1 standard
- 4 undersized (-0.025mm to -0.1mm in 0.025mm steps)
- 4 oversized (+0.025mm to +0.1mm in 0.025mm steps)
- Housing Components: All structural elements
- Output Components: Output disc and related hardware
- Pre-Oriented STL Files: For optimal 3D print performance
- Bill of Materials (BOM): Complete hardware list

Premium Package (STEP Files) Includes everything in the Standard Package, plus:

- STEP Files: For CAD modification
- Complete Assembly Model: For reference (hardware not included)

Key Features

- Reduction Ratio: 20:1
- Form Factor: Compact and desktop-ready
- Precision Fit: 9-disc variations for ideal tolerance
- Smooth Motion: Reduced vibration via cycloidal geometry
- Efficient Design: High torque in a small package

Warranty & Disclaimer

This is a DIY assembly product. Sweep Dynamics provides files as-is. Final performance depends on print quality, fit, lubrication, and assembly precision. Please test thoroughly before using in critical applications.

3D Printing Settings Table

Parameter	Value	Notes
Layer Height	0.20 mm	Balance of detail and strength
Initial Layer Height	0.25 mm	Improves bed adhesion
Wall Count	3-4	Increases structural integrity
Top/Bottom Layers	7 / 7	Ensures solid surfaces
Infill Density	25-30%	Sufficient for mechanical loads
Infill Pattern	Gyroid or Grid	Strength-to-weight ratio
Infill/Wall Overlap	15%	Enhances bonding
Sparse Infill Area	Min. 15 mm^2	Prevents weak spots
Support Structure	Yes	Required for overhangs
Support Density	15%	Material-efficient
Interface Layers	2	Smooth contact surface
Z Distance	0.2 mm	Clean support separation

Size Selection Guide

Type	Tolerance	Use When...
Standard	+/-0.000 mm	Start here for most prints
Undersized	-0.025 to -0.1 mm	If fit is too tight
Oversized	+0.025 to +0.1 mm	If fit is too loose